

Werner Bisschoff

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Enterprise Systems & Automation Architect

Professional Summary

Strategic Enterprise Software Architect with 5+ years of experience engineering robust business automation platforms, custom ERP extensions, and scalable API layers. Expert in the Frappe/ERPNext ecosystem, asynchronous workflow systems, and cross-platform mobile application deployment (Expo/React Native). Proven track record of translating complex real-world operations into scalable, type-safe data schemas governed by strict Spec-Driven Development (SDD) paradigms.

Experience

Principal Systems Architect , Cape Town Divergent Tabletop | Jul 2025

- Implemented PostgreSQL Row-Level Security (RLS) and SET LOCAL transaction-scoped context injection for multi-tenant isolation.
- Designed atomic asynchronous flush strategy using in-memory GenServer buffers, reducing database writes by 85%.
- Utilized Ecto.Multi atomic chains and Oban transactional workers for safe multi-stage backend edits.

Full-Stack ERPNext & Automation Engineer , Cape Town FARO Africa | Aug 2024 – Nov 2025

- Refactored and extended ERPNext via Python and JavaScript server hooks, optimizing pricing matrices and BI reporting.
- Architected migration of internal Retool workflows to Expo/React Native, reducing vendor licensing overhead.
- Built custom NFC scanner utilities within Expo for instantaneous physical inventory syncs.
- Provisioned AWS infrastructure with Pulumi, deploying Inngest event-driven functions and PayloadCMS.

Enterprise Infrastructure & Tooling Engineer , Re-Ingenics Digital GmbH (through ViVa Outsourcing) | Mar 2023 – May 2024
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- Built modular Python utilities and automation scripts for internal configuration schema validation.
- Designed data handling pipelines ensuring transfer consistency across runtime platforms.
- Developed Python hardware mocks enabling faster development cycles with fewer hardware dependencies.

Junior Lecturer , Potchefstroom North-West University | Feb 2020 – Dec 2020

- Lectured Python and C++ programming for first-year IT students.

Technical Application Scope

Primary	Python ERPNext/Frappe Framework Expo (React Native) Inngest
Secondary	AWS Pulumi (IaC) PostgreSQL (RLS) Docker
Foundational	SQL Object-Oriented Design Spec-Driven Development
Other	C++ FreeRTOS (Low-level integration support)

Education

B.Eng. Computer and Electronic Engineering North-West University | 2020

Focus on embedded systems, software engineering, and electronic design · Developed data analysis pipeline with Python/Pandas (linear regression, ML); built NFC payment emulator (Kotlin/Android)